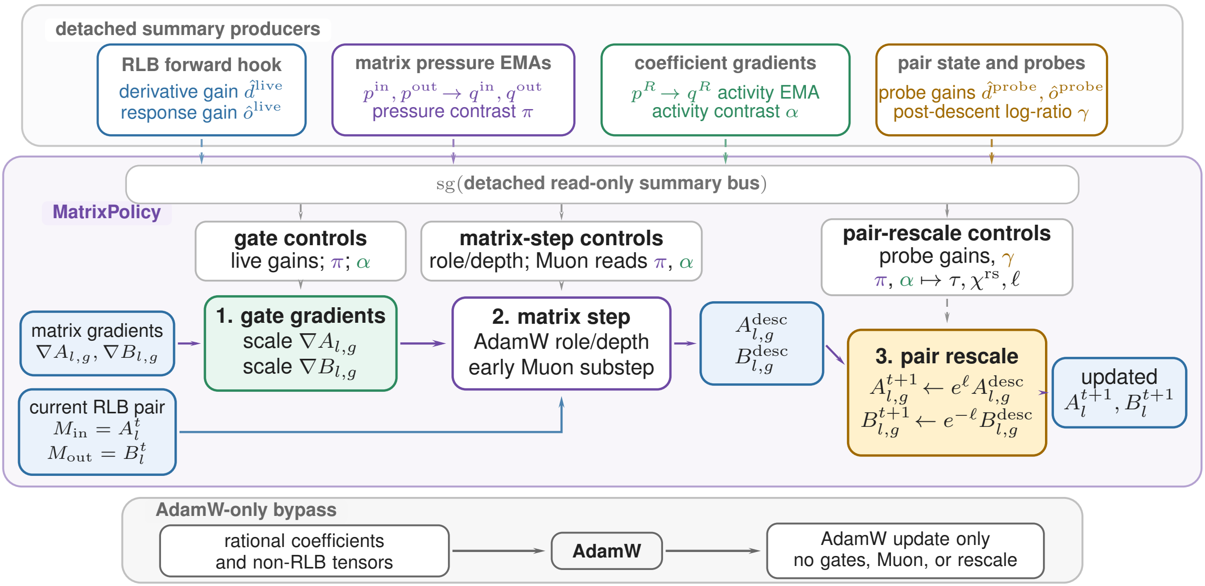

detached summary producers
RLB forward hook
derivative gain $\hat{d}^{\text{live}}$
response gain $\hat{o}^{\text{live}}$
matrix pressure EMAs
$p^{\text{in}}, p^{\text{out}} \to q^{\text{in}}, q^{\text{out}}$
pressure contrast π
coefficient gradients
$p^R \to q^R$ activity EMA
activity contrast α
pair state and probes
probe gains $\hat{d}^{\text{probe}}, \hat{o}^{\text{probe}}$
post-descent log-ratio γ
sg(detached read-only summary bus)
MatrixPolicy
gate controls
live gains; π; α
matrix-step controls
role/depth; Muon reads π, α
pair-rescale controls
probe gains, γ
$\pi, \alpha \mapsto \tau, \chi^{\text{rs}}, \ell$
matrix gradients
$\nabla A_{l,g}, \nabla B_{l,g}$
1. gate gradients
scale $\nabla A_{l,g}$
scale $\nabla B_{l,g}$
2. matrix step
AdamW role/depth
early Muon substep
$A^{\text{desc}}_{l,g}$
$B^{\text{desc}}_{l,g}$
3. pair rescale
$A^{t+1}_{l,g} \leftarrow e^{\ell} A^{\text{desc}}_{l,g}$
$B^{t+1}_{l,g} \leftarrow e^{-\ell} B^{\text{desc}}_{l,g}$
updated
A^{t+1}_l, B^{t+1}_l
current RLB pair
$M_{\text{in}} = A^t_l$
$M_{\text{out}} = B^t_l$
AdamW-only bypass
rational coefficients
and non-RLB tensors
AdamW
AdamW update only
no gates, Muon, or rescale